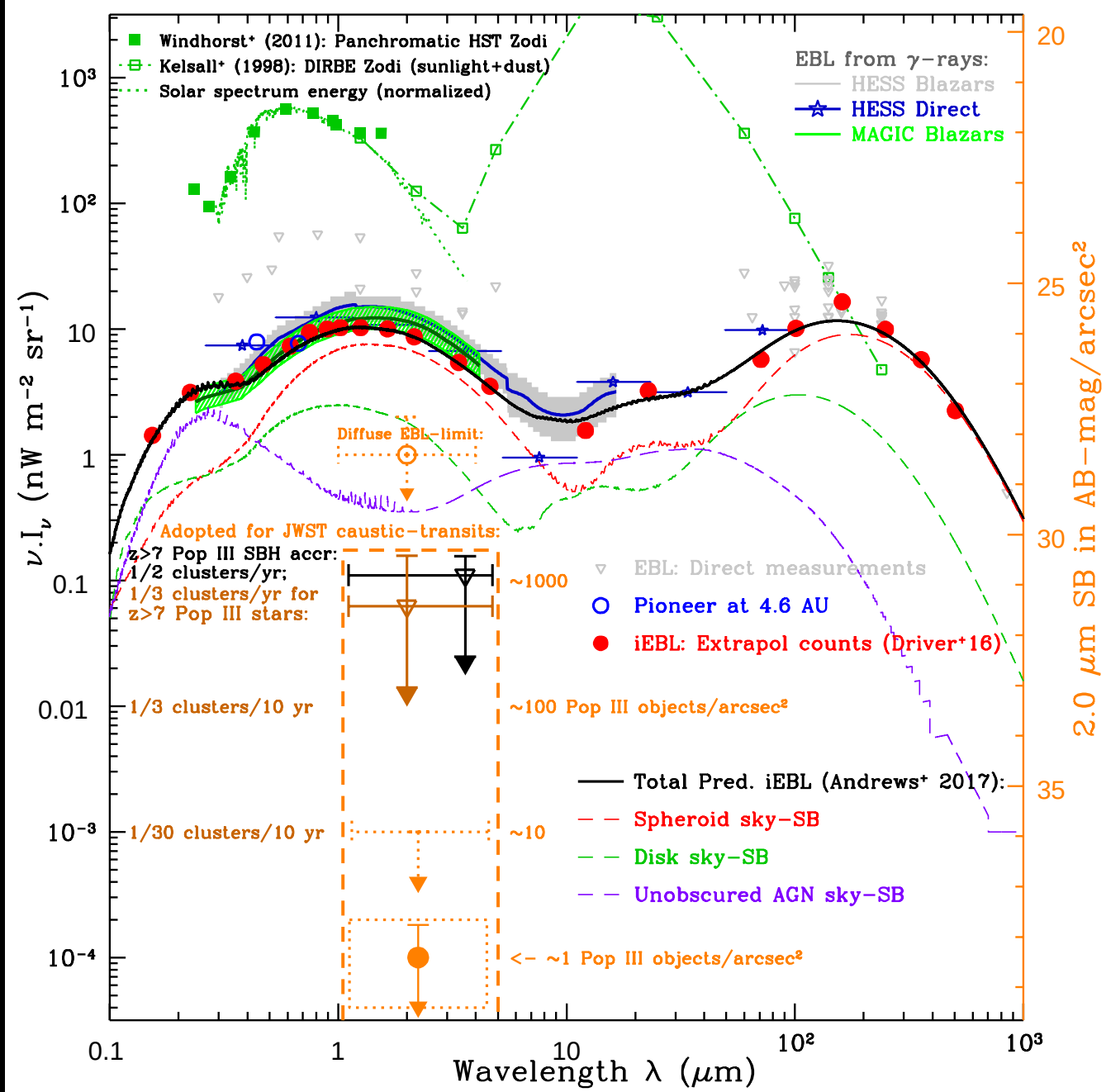




Most important JWST work:

Individual lensed stars at $z \sim 0.7-6.2$!

But Pop III stars must be some part of the diffuse EBL: their 1–4 μ m SB $\lesssim 0.1$ nW/m²/sr or $\gtrsim 31$ mag/arcsec²:

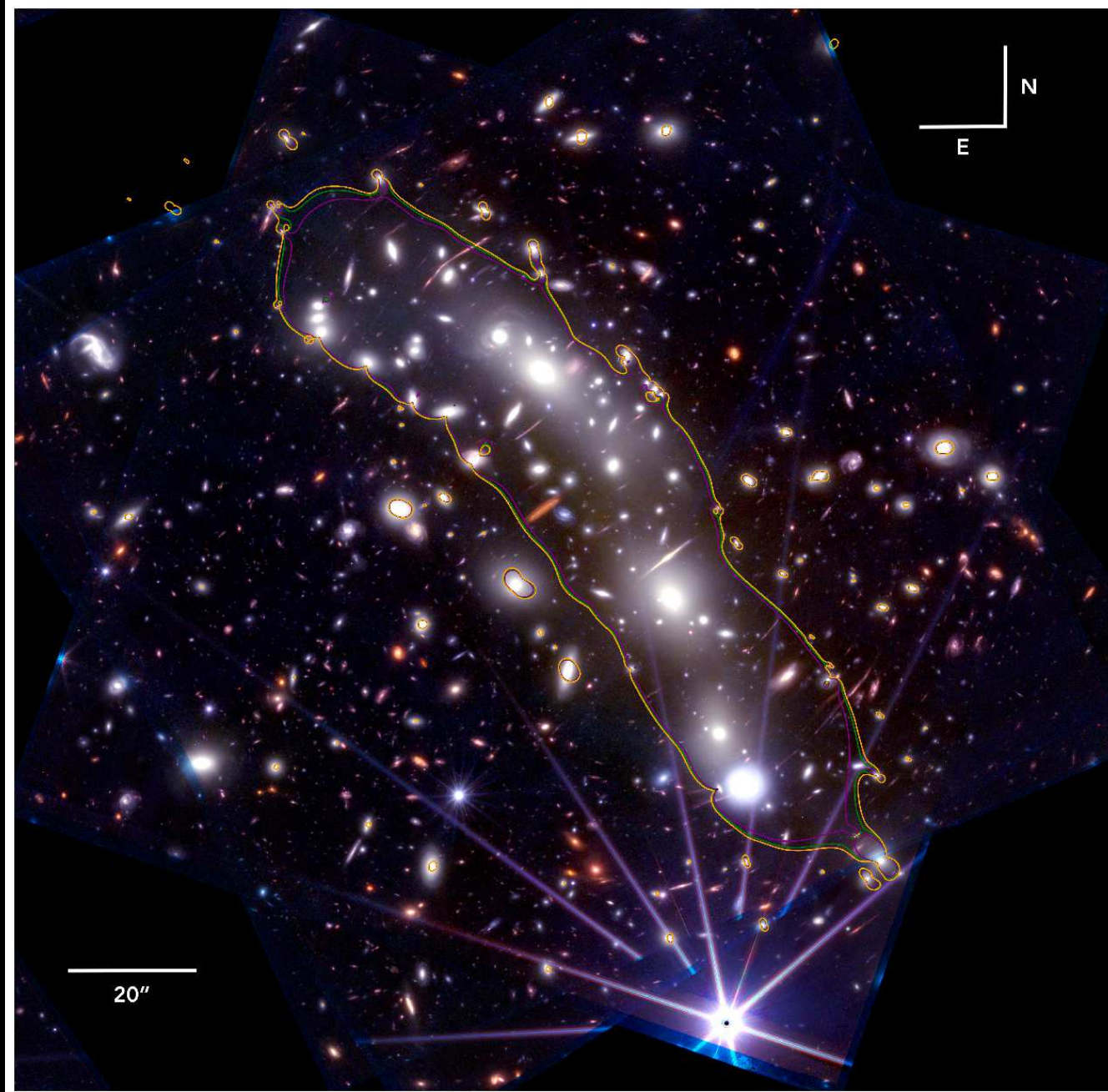
- 1) possibly from Pop III stars at $z \gtrsim 7-17$, and/or their
- 2) stellar-mass BH accretion disks at $z \gtrsim 7$ (Kashlinsky⁺ 2015, Mitchell-Wynne⁺ 2016, Windhorst⁺ 2018).

What biggest unanswered question?:

- No lensed Pop III stars detected at $z \gtrsim 6.2$! Why??? Why not yet?

Biggest barriers to making progress?:

- Pop III stars & their BH accretion disks should be visible to JWST through caustic transits at AB $\lesssim 31$.
- Pop III RGB and AGB stars are better detectable (*i.e.*, larger; microlensing!) than ZAMS stars.
- Pop III star detection: need to monitor 30–60 best lensing clusters for caustic transits for $\sim 5-10$ yrs.



Most important JWST work:

Individual lensed stars at $z \sim 0.7-6.2$!

MACS0416: New limits (Perivoltis⁺ 2026; tightened W18 prediction):

- $L_{caustic}(z=7-17) \simeq 293 \pm 33''$.
- 3 epochs months to a year apart.
- Transverse $v_T \simeq 1000-2000$ km/s.

$\Rightarrow$ Pop III SB(K) $\gtrsim 33$ mag arcsec⁻² :
 $\lesssim 100$ Pop III * (30–100 $M_\odot$) / ''².

What biggest unanswered question?:

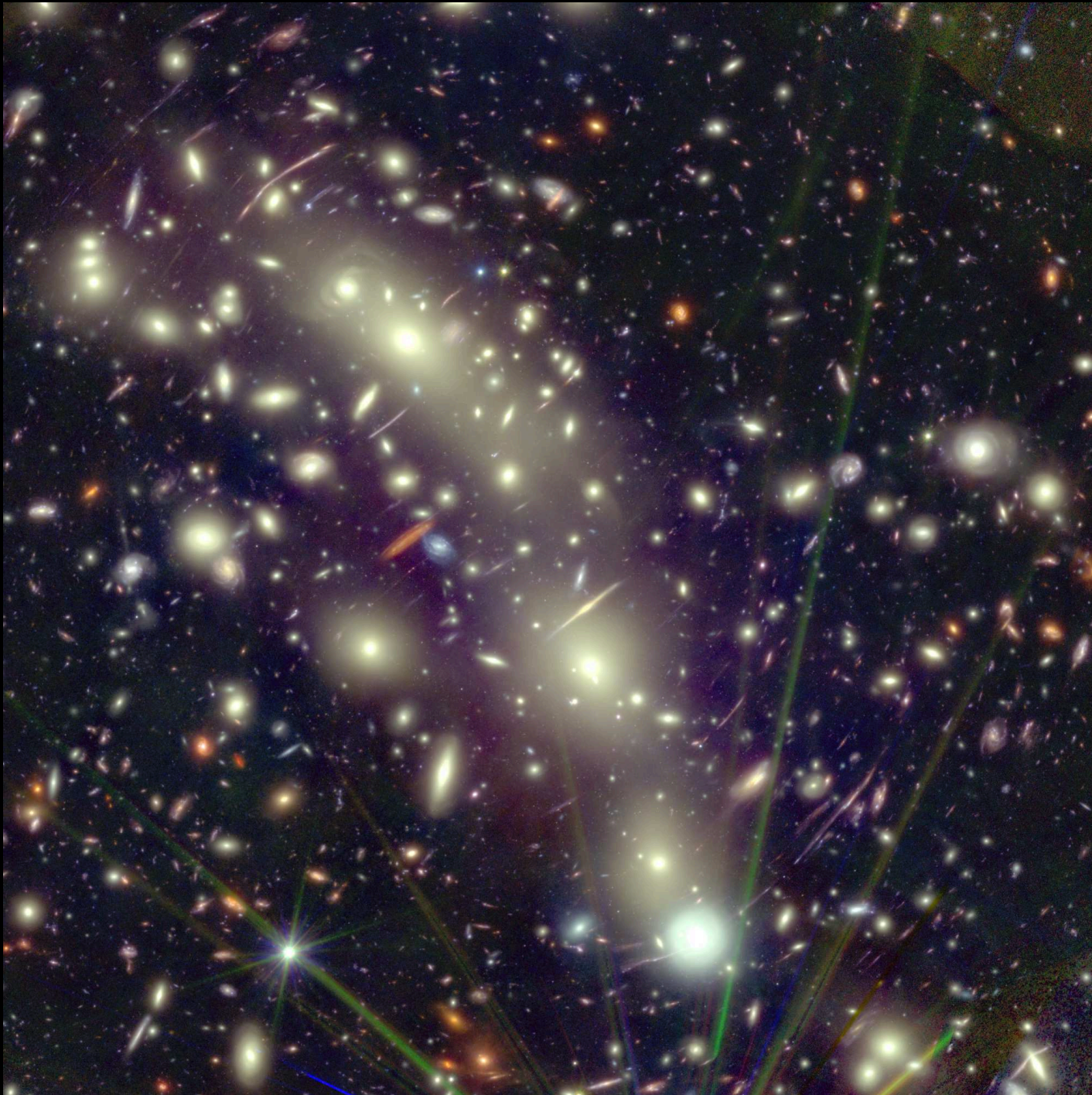
• No lensed Pop III stars detected at $z \gtrsim 6.2$! Why??? Why not yet?

Biggest barriers to making progress?:

- Need much more $L_{caustic}$ length.
- Select likely highest v_T velocities.
- Monitor few times/yr (in 2 filters), depending on ICL at $z \gtrsim 7$ CC.
- Then many-filter TOO follow-up for $z \gtrsim 7$ Pop III star caustic transits.

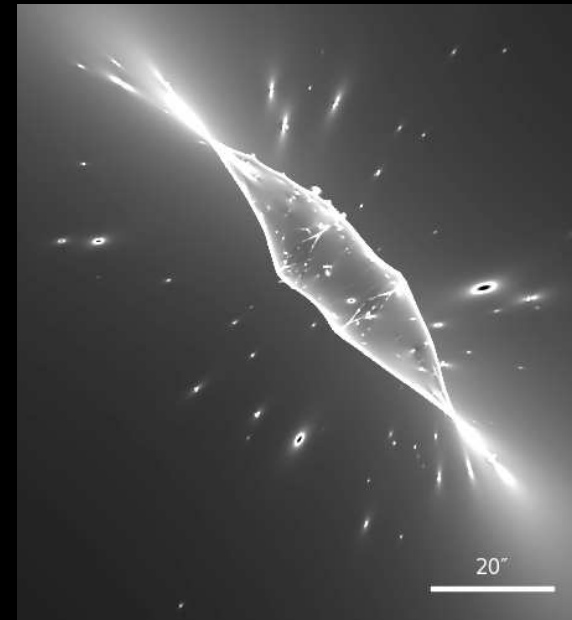
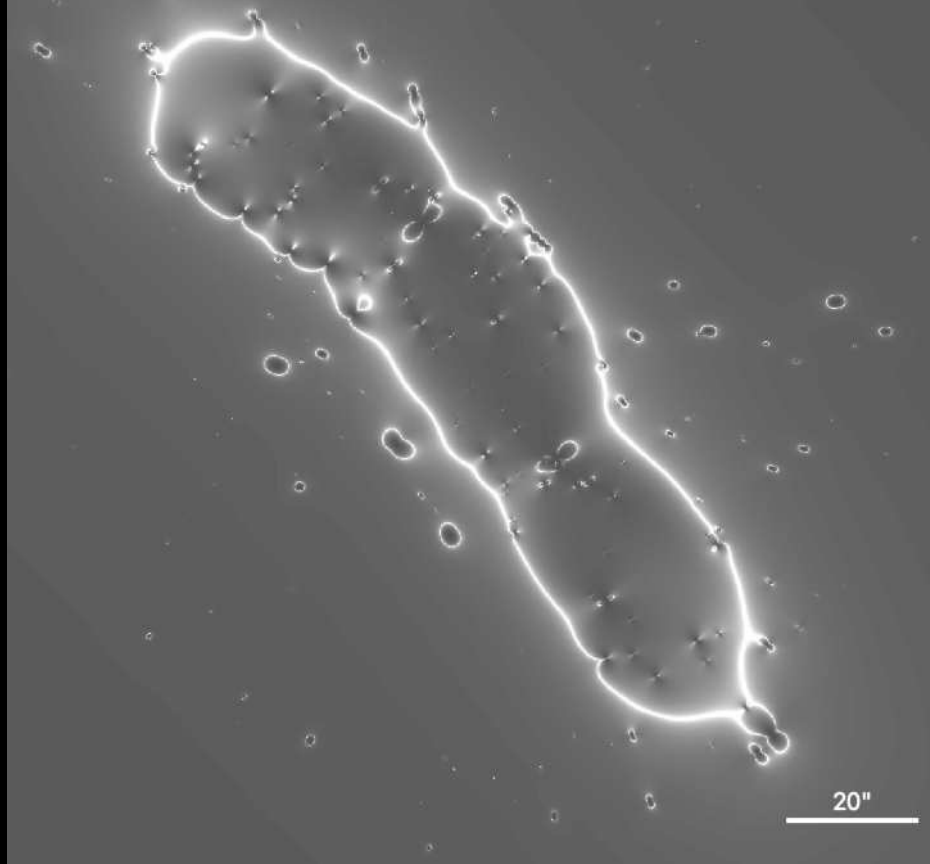
- Pop III stars & their BH accretion disks should be visible to JWST through caustic transits at $AB \lesssim 31$.
- Pop III RGB and AGB stars are better detectable (*i.e.*, larger; microlensing!) than ZAMS stars.
- Pop III star detection: need to monitor 30–60 best lensing clusters for caustic transits for $\sim 5-10$ yrs.

SPARE CHARTS (for Th/Fr VENUS session)



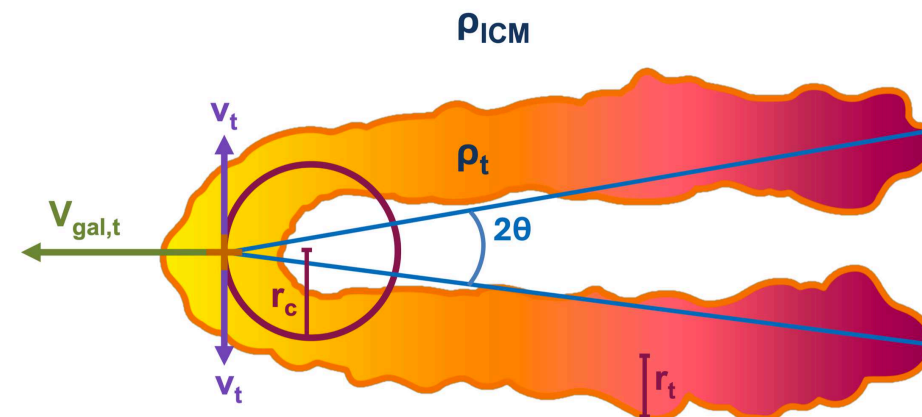
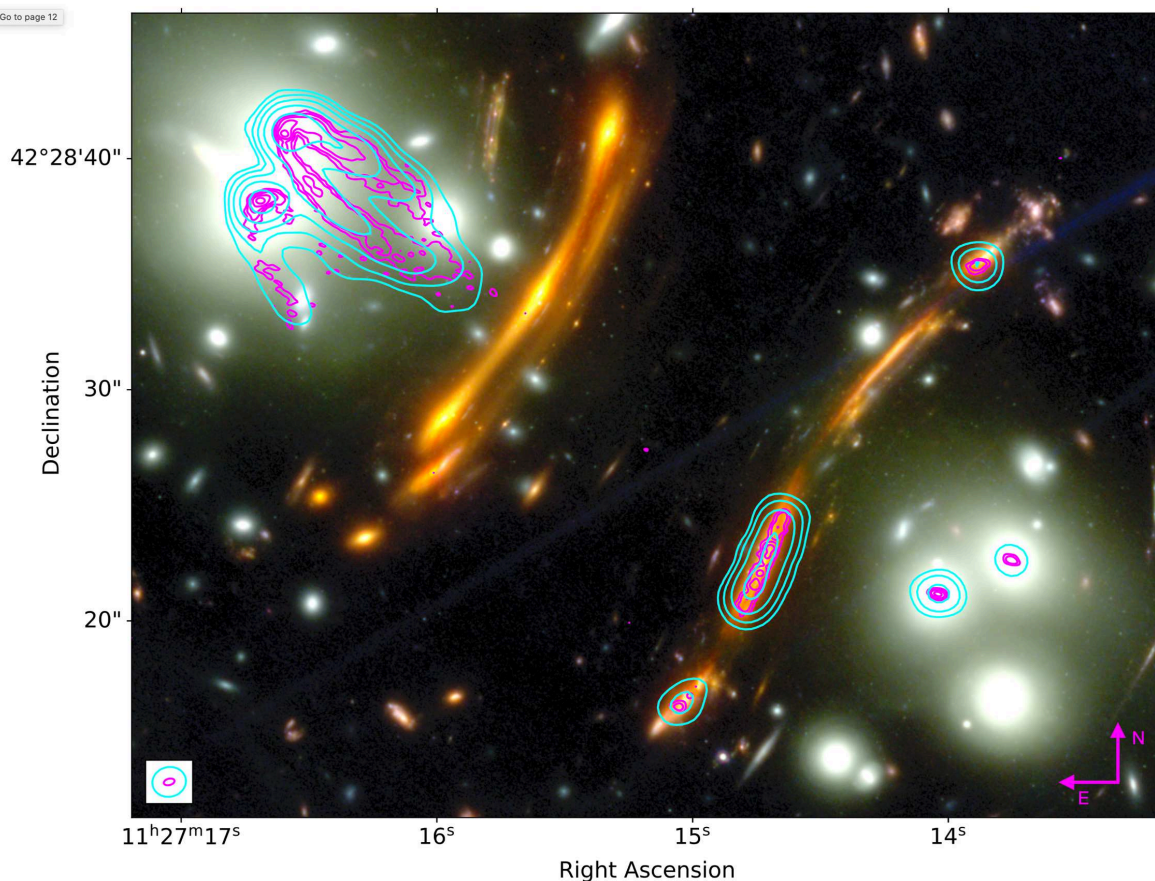
500 hrs HST+JWST: 45 filters ($0.2\text{--}5.0\mu\text{m}$), lensing cluster MACS0416:

- HST darkest skies ($10\text{--}10^3\times$ darker) + JWST's dark skies ($10^3\text{--}10^5\times$ darker than ground based):



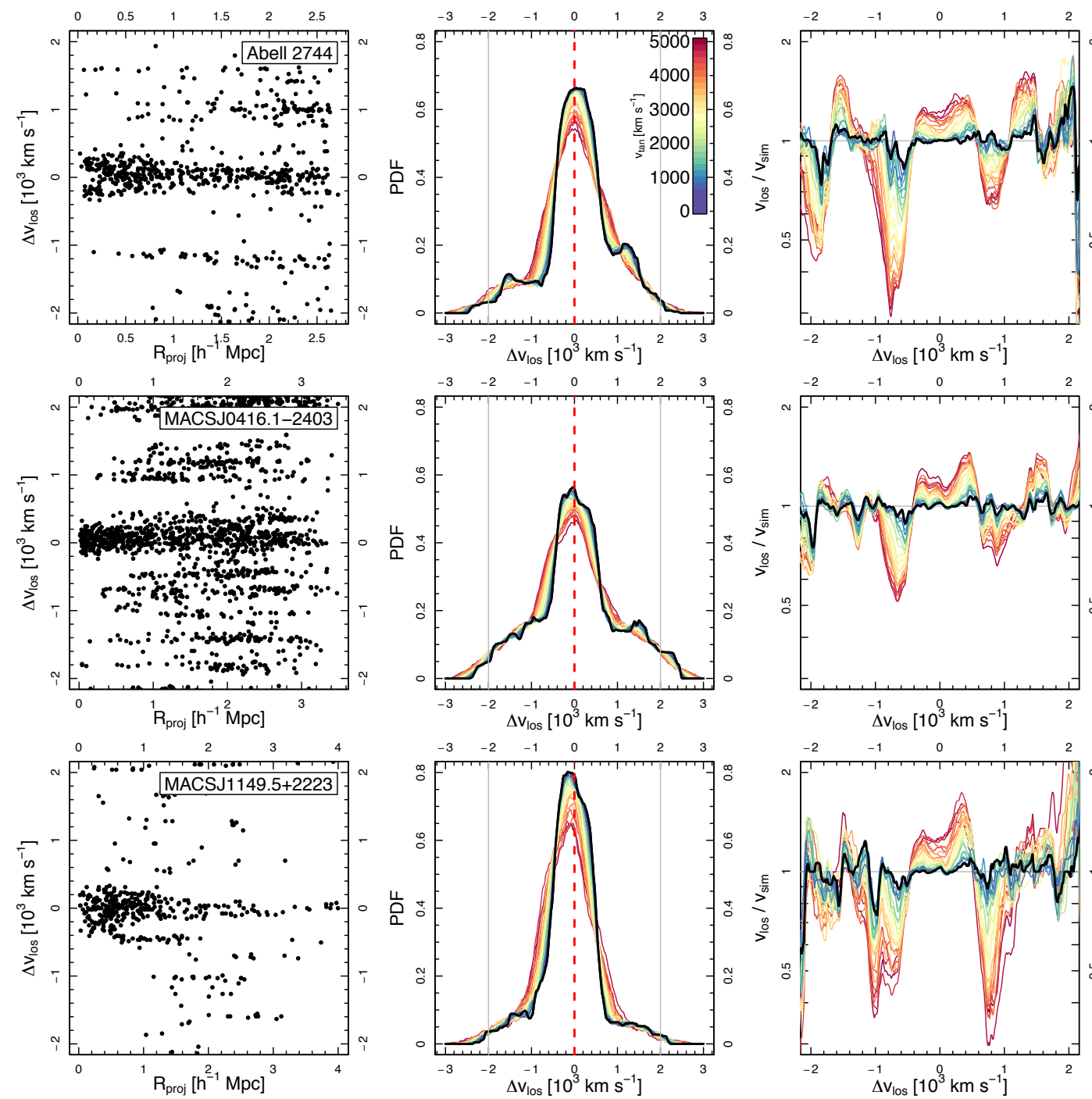
For source at $z=10$, critical curves for HFF cluster MACS 0416 at $z \simeq 0.397$ [LEFT], and main cluster caustics [in source plane; RIGHT – note scales].

- Transverse cluster (sub-component) velocities: $v_T \simeq 1000\text{--}2000$ km/s (Kelly⁺ 2018; Windhorst⁺ 2018; Wilde, Frye⁺ 2026).
- Main caustic magnification: $\mu \simeq 10.(d_{caustic}/")^{-1/2}$. For Pop III objects at $z \gtrsim 7$ with $1\text{--}100 R_\odot$, μ can be $\gtrsim 10^4\text{--}10^5$ for $\lesssim 0.4$ year!
- Must use clusters with minimal ICL near the critical curves, since ICL microlensing will dilute the main caustics (Diego⁺ 2018, ApJ, 857, 25).



Planck cluster G165 ($z=0.348$) has two head-tail radio sources:

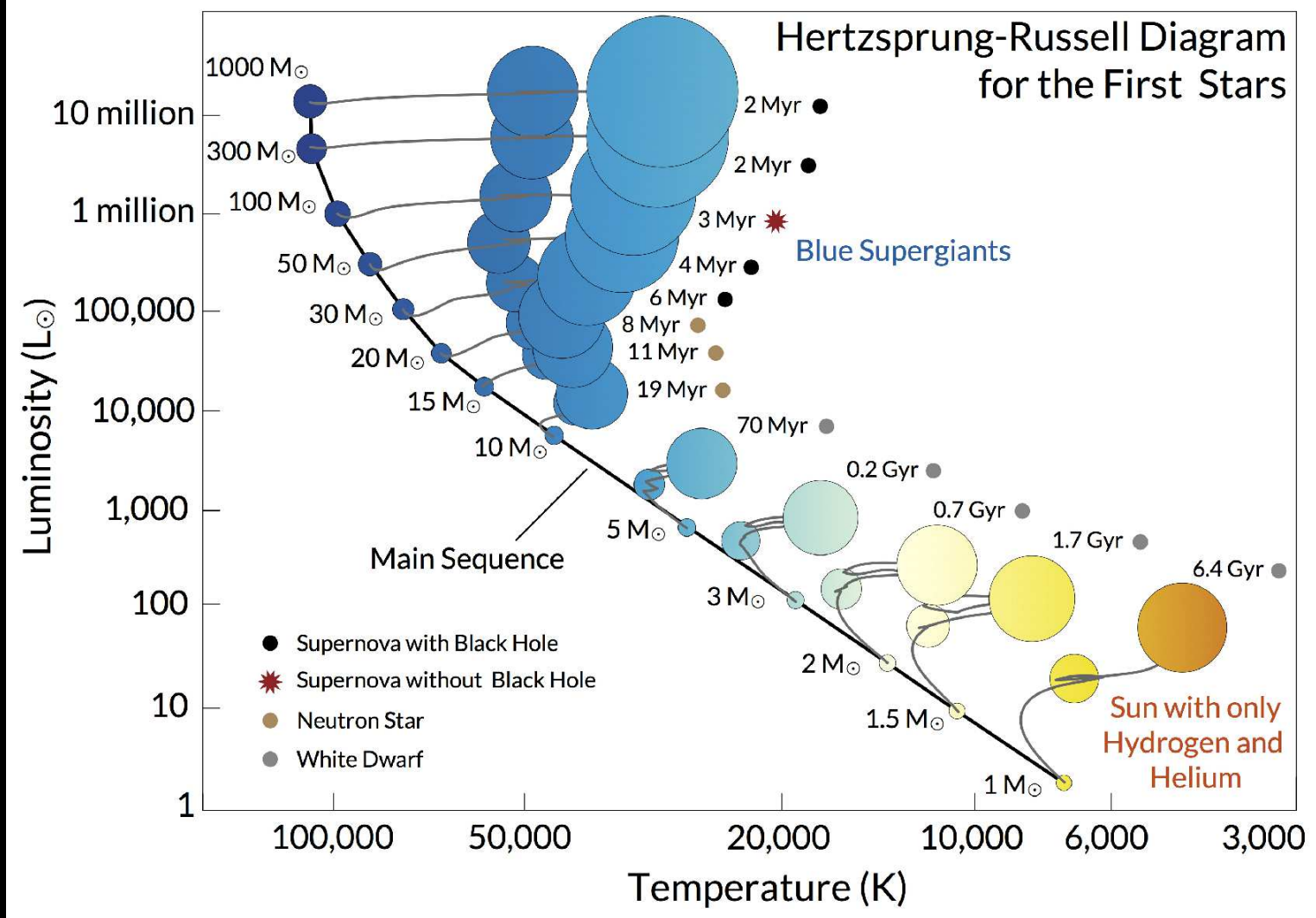
- VLA 6 & 15 GHz images $\Rightarrow$ transverse cluster (component) velocities:
- $v_T \simeq 2200 \pm 1500$ km/s from nonrelativistic hydrodynamic flow model.
- $v_T \simeq 1800 \pm 650$ km/s from Mach cone model (Wilde, Frye, et al. 2026).
- $v_T \sim 1000\text{--}2000$ km/s are good numbers to use for (merging) clusters!



Trumpet diagrams for JWST lensing clusters from ground-based spectroscopic $N(z)$ (Windhorst⁺ 2018):

- 1) Add random *space* velocity v_{sp} to clusters.
- 2) Projected v_T must be $\lesssim 1900$ km/s for v_{sp} not to unduly disturb radial $N(z)$.
- 3) Best clusters (Bullet) for caustic transits can have $v_T \lesssim 2700$ km s⁻¹.

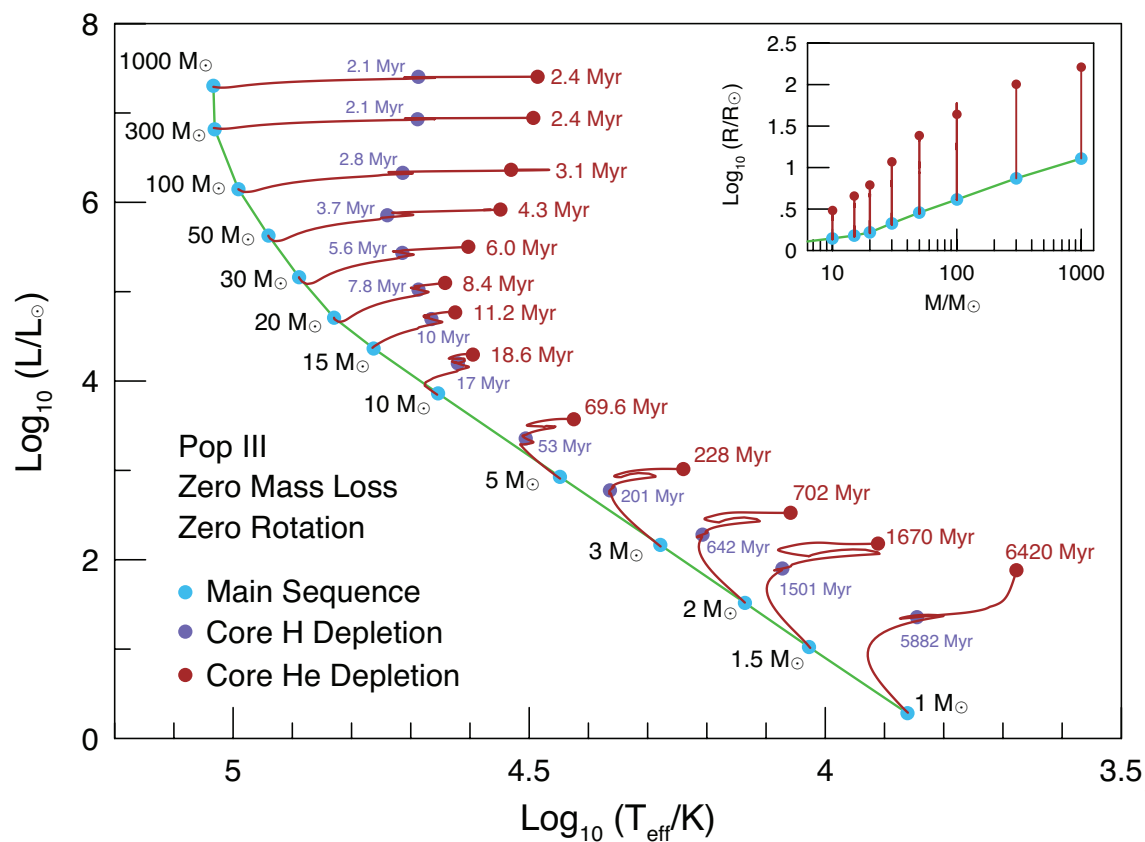
• JWST must monitor such clusters during its lifetime for caustic transits.



Pop III star HR-diagram: MESA stellar evolution models for $z=0.0$ $Z_{\odot}$.

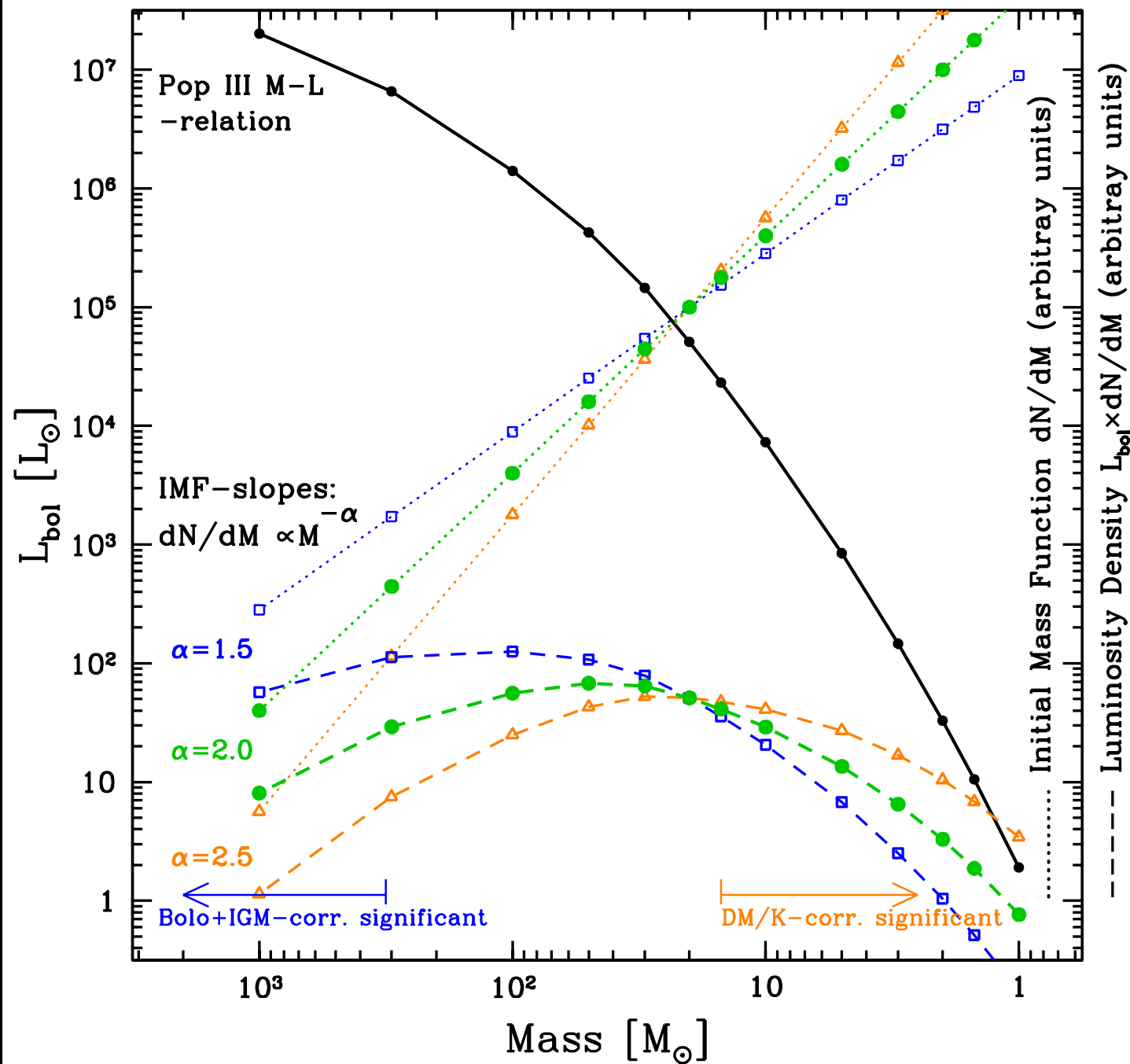
(Windhorst, Timmes, Wyithe et al. 2018, ApJS, 234, 41): $\mu \gtrsim 8-10$ mag could make Pop III stars temporarily visible.

- Critical point: 30–1000 $M_{\odot}$ Pop III stars ($Z=0.00$ $Z_{\odot}$) live $\sim 10\times$ shorter than 2–5 $M_{\odot}$ Pop III stars in their AGB stage.
- Hence, 2–5 $M_{\odot}$ AGB companion stars can feed the LIGO-mass BHs left over from $M \gtrsim 30$ $M_{\odot}$ Pop III stars (assuming binaries in 2nd generation).

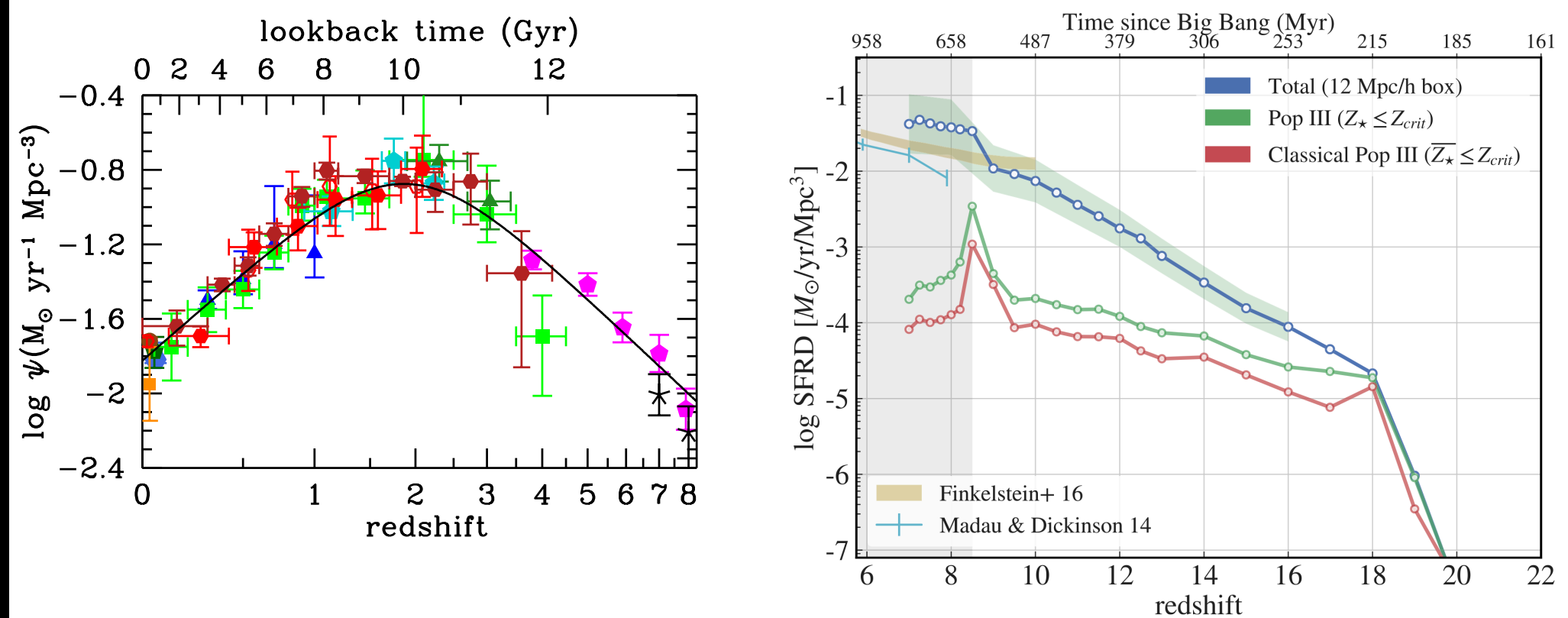


Windhorst⁺ (2018, ApJS, 234, 41):

- Multicolor accretion-disk models for stellar-mass black holes [RIGHT]: For $M_{BH} \simeq 5\text{--}700 M_{\odot}$, accretion disk radii and luminosities are similar to those of Pop III AGB stars, when the BH is fed by a Roche lobe-filling lower-mass companion star on the AGB (which live $\gtrsim 10\times$ longer!).
- Assumes 2nd generation O-stars have high enough Fe/H ($\gtrsim 10^{-4} Z_{\odot}$) that $2\text{--}5 M_{\odot}$ AGB companion stars exist and feed these LIGO-mass BHs.
- This may make stellar-mass black hole accretion disks at least as likely to be seen via caustic transits as the Pop III stars themselves ($\mu \gtrsim 10^4$).



Mass-Luminosity relation for zero metallicity Pop III MESA models:
For range of IMF slopes, most Pop III star sky-SB comes from 20–300 $M_{\odot}$!



Adopted cosmic star-formation rate (SFR) at $z \gtrsim 7$:

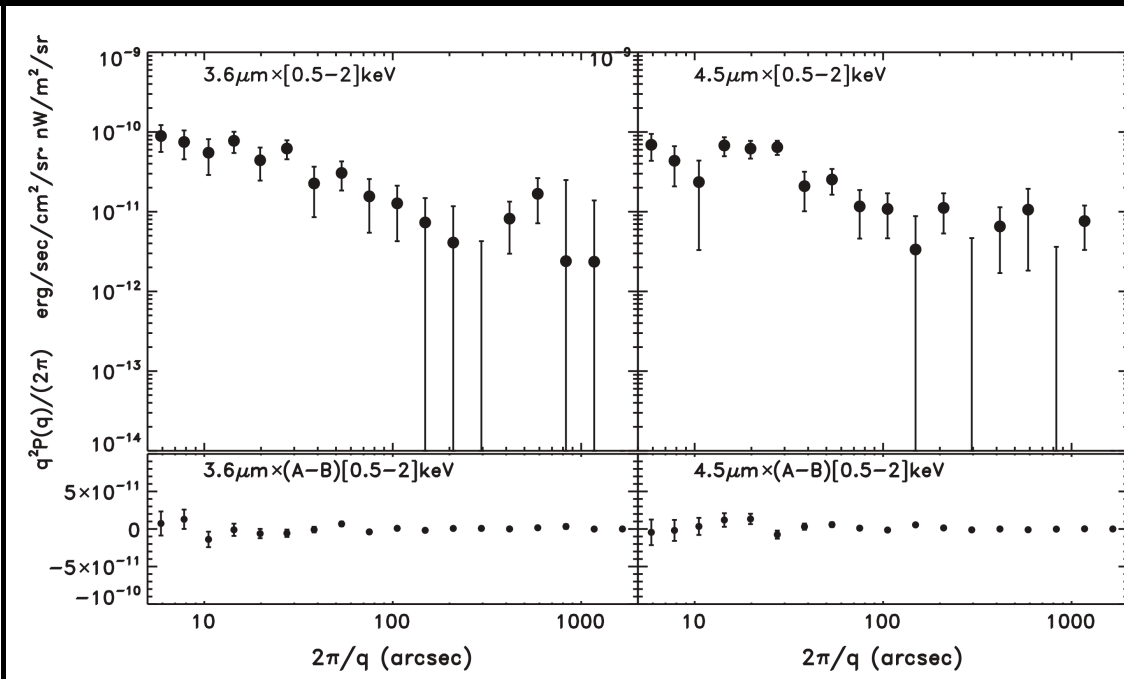
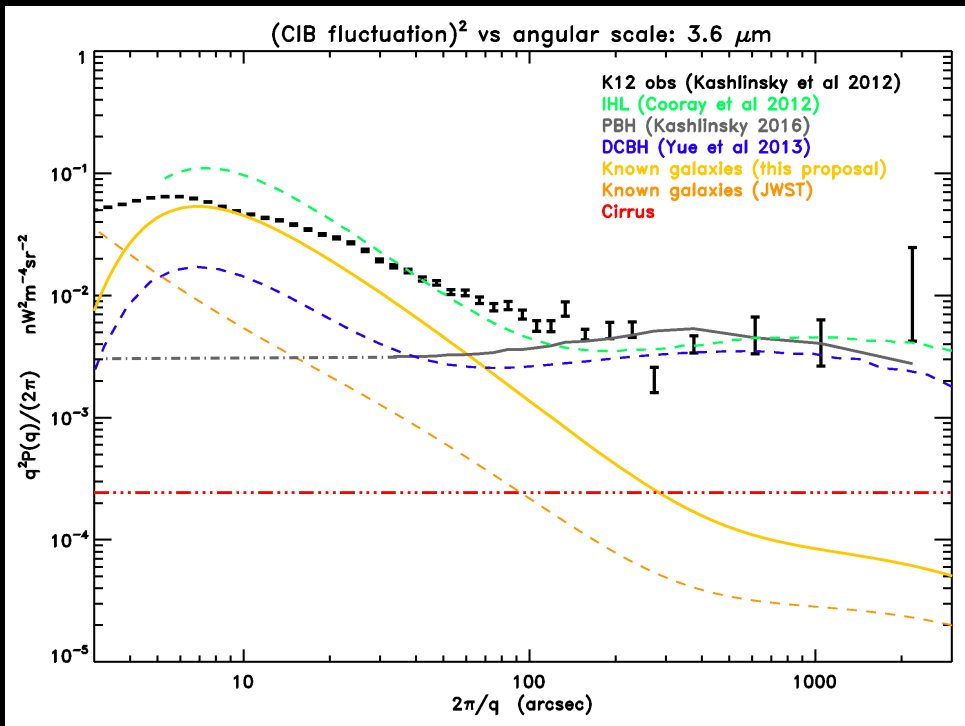
[LEFT] Observed SFH (Madau & Dickinson; 2014 ARAA, 52, 415);

[RIGHT] RAMSES models (*e.g.*, Sarmento et al. 2018, ApJ, 854 75).

$\Rightarrow$ Adopt this SFR from $z \simeq 17$ to $z \simeq 7$, implying at the lowest masses:

- Metallicity increases from ~ 0 at $z \simeq 18$ to $\lesssim 10^{-3}$ solar at $z \simeq 7$.
- Integrated SFR from $z \gtrsim 7$ has sky-SB $\gtrsim 31 \text{ K-mag/arcsec}^{-2}$ (Windhorst et al. 2018), similar to the $3.6 \mu\text{m}$ CIB sky-SB possibly from ** BH's.

(3a) Limits to Pop III Sky-SB: First (Stellar-Mass?) Black Holes



[LEFT] Object-free Spitzer 3.6 μm power-spectrum constrains noise fluctuation models (Cappelluti et al. 2017; Kashlinsky et al. 2012, 2015, 2018):

Explainable by: Primordial black hole or Direct-collapse black hole models.

[RIGHT] Spitzer–Chandra cross-corr spectrum (Mitchell-Wynne et al. 2016):

- $z \gtrsim 7$ objects have sky-SB fainter than 31 mag/arcs², plus likely a (stellar mass) black hole X-ray component. (Kashlinsky⁺ 2018; Windhorst⁺ 2018, ApJ, 234, 41).

Table 1. Adopted Pop III Star Physical Parameters from MESA models^a

Mass	Age	T_{eff}	$\log R$	$\log L_{bol}$	T_{eff}	$\log R$	$\log L_{bol}$	Age	T_{eff}	$\log R$	$\log L_{bol}$	Age	Time ^b
	Pre-MS	— at ZAMS —			— at Hydrogen-depletion —				— at Helium-depletion —				AGB-MS
($M_{\odot}$)	(Myr)	(K)	($R_{\odot}$)	($L_{\odot}$)	(K)	($R_{\odot}$)	($L_{\odot}$)	Myr	(K)	($R_{\odot}$)	($L_{\odot}$)	Myr	(Myr)
1.0	9.28	7.266e3	-0.0581	0.2825	6.999e3	0.5119	1.3576	5882	— ^c	—	—	6420	538
1.5	6.11	1.065e4	-0.0203	1.0227	1.181e4	0.3292	1.9015	1501	8.149e3	0.7913	2.1804	1670	169
2.0	3.02	1.367e4	0.0108	1.5177	1.611e4	0.2498	2.2815	642	1.145e4	0.6685	2.5249	702	60
3.0	1.38	1.899e4	0.0487	2.1654	2.311e4	0.1843	2.7770	201	1.736e4	0.5510	3.0138	228	27
5.0	0.56	2.805e4	0.0911	2.9274	3.206e4	0.1903	3.3581	53	2.658e4	0.4608	3.5732	70	17
10	0.23	4.508e4	0.1462	3.8618	4.174e4	0.3807	4.1972	17	3.938e4	0.4811	4.2968	19	1.6
15	0.13	5.789e4	0.1803	4.3647	4.624e4	0.5401	4.6937	10	4.215e4	0.6581	4.7691	11	0.8
20	0.09	6.754e4	0.2183	4.7082	4.864e4	0.6612	5.0240	7.8	4.386e4	0.7879	5.0975	8.4	0.6
30	0.05	7.737e4	0.3270	5.1619	5.180e4	0.8120	5.4347	5.6	4.006e4	1.0688	5.5016	6.0	0.5
50	0.03	8.713e4	0.4570	5.6283	5.490e4	0.9722	5.8562	3.7	3.536e4	1.3862	5.9200	4.3	0.5
100	0.02	9.796e4	0.6147	6.1470	5.173e4	1.2610	6.3303	2.8	3.392e4	1.6437	6.3627	3.1	0.3
300	0.02	1.074e5	0.8697	6.8172	4.882e4	1.6111	6.9301	2.1	3.165e4	2.0041	6.9631	2.4	0.3
1000	0.02	1.080e5	1.1090	7.3047	4.807e4	1.8740	7.4288	2.1	3.122e4	2.2119	7.3549	2.4	0.3

Windhorst, Timmes, Wyithe et al. (2018, ApJS, 234, 41):

- 30–1000 $M_{\odot}$ Pop III stars ($Z=0.00$ $Z_{\odot}$) live $\sim 10\times$ shorter than 2–5 $M_{\odot}$ Pop III stars in their AGB stage.
- Hence, 2–5 $M_{\odot}$ AGB companion stars can feed the LIGO-mass BHs left over from $M \gtrsim 30$ $M_{\odot}$ Pop III stars (assuming binaries in 2nd generation).

Table 2. Implied ZAMS Pop III Star Observational Parameters Relevant to Caustic Transit Calculations

Mass ^a	T_{eff}^b	Radius ^c	L_{bol}^d	M_{bol}^e	Bolo+IGM+K-corr ^f			ZAMS m_{UV}^g			t_{rise}^h	transit ⁱ
ZAMS		— at ZAMS —			z=7	z=12	z=17	z=7	z=12	z=17	caust	rate
($M_{\odot}$)	(K)	($R_{\odot}$)	($L_{\odot}$)	(AB)	(AB-mag)			(AB-mag)			(hr)	(/cl/yr)
1.0	7.266e3	0.87	1.92	+4.03	+4.44	+3.13	+2.61	57.71	57.74	58.07	0.17	8×10^5
1.5	1.065e4	0.95	10.5	+2.18	+1.45	+0.42	−0.06	52.87	53.18	53.55	0.18	1.1×10^4
2.0	1.367e4	1.03	32.9	+0.95	+0.30	−0.59	−1.06	50.49	50.93	51.31	0.20	1.5×10^3
3.0	1.899e4	1.12	146.	−0.67	−0.51	−1.26	−1.72	48.06	48.64	49.03	0.22	182.
5.0	2.805e4	1.23	846.	−2.58	−0.70	−1.35	−1.80	45.96	46.65	47.04	0.24	29.1
10	4.508e4	1.40	7.28e3	−4.91	−0.22	−0.79	−1.23	44.10	44.88	45.27	0.27	5.70
15	5.789e4	1.51	2.32e4	−6.17	+0.23	−0.30	−0.75	43.30	44.10	44.50	0.29	2.78
20	6.754e4	1.65	5.11e4	−7.03	+0.56	+0.04	−0.40	42.77	43.59	43.99	0.32	1.74
30	7.737e4	2.12	1.45e5	−8.16	+0.88	+0.36	−0.08	41.95	42.78	43.17	0.41?	0.82?
50	8.713e4	2.86	4.25e5	−9.33	+1.17	+0.66	+0.22	41.08	41.91	42.31	0.55*	0.37*
100	9.796e4	4.12	1.40e6	−10.63	+1.47	+0.96	+0.52	40.08	40.91	41.31	0.80*	0.15*
300	1.074e5	7.41	6.56e6	−12.30	+1.71	+1.21	+0.77	38.64	39.48	39.88	1.43*	0.039*
1000	1.080e5	12.9	2.02e7	−13.52	+1.72	+1.22	+0.78	37.44	38.28	38.68	2.48*	0.013*

- If $M \gtrsim 30 M_{\odot}$ Pop III ZAMS stars have $\mu \gtrsim 10^4$ – 10^5 during caustic transits, they could be detectable for months to $\text{AB} \lesssim 29$ mag with JWST.
- Expect $\lesssim 1$ caustic transit/yr at $z \gtrsim 7$ when JWST monitors $\gtrsim 30$ clusters.

Table 3. Implied Red Giant Branch Pop III Star Observational Parameters Relevant to Caustic Transit Calculations

Mass ^a	T_{eff}^b	Radius ^c	L_{bol}^d	M_{bol}^e	Bolo+IGM+K-corr ^f			Giant Branch m_{UV}^g			t_{rise}^h	transit ⁱ
GB	— at Hydrogen-depletion —				z=7	z=12	z=17	z=7	z=12	z=17	caust	rate
($M_{\odot}$)	(K)	($R_{\odot}$)	($L_{\odot}$)	(AB)	(AB-mag)			(AB-mag)			(hr)	(/cl/yr)
1.0	6.999e3	3.25	22.8	+1.35	+4.83	+3.48	+2.96	55.42	55.41	55.73	0.63	9×10^4
1.5	1.181e4	2.13	79.7	−0.01	+0.91	−0.06	−0.53	50.13	50.51	50.88	0.41	1.0×10^3
2.0	1.611e4	1.78	191.	−0.96	−0.19	−1.01	−1.47	48.08	48.60	48.99	0.34	175.
3.0	2.311e4	1.53	598.	−2.20	−0.69	−1.39	−1.84	46.35	46.99	47.38	0.30	39.8
5.0	3.206e4	1.55	2.28e3	−3.66	−0.63	−1.25	−1.70	44.95	45.67	46.07	0.30	11.8
10	4.174e4	2.40	1.57e4	−5.75	−0.34	−0.92	−1.36	43.15	43.91	44.31	0.46	2.33
15	4.624e4	3.47	4.94e4	−6.99	−0.18	−0.74	−1.19	42.06	42.84	43.24	0.67?	0.87?
20	4.864e4	4.58	1.06e5	−7.82	−0.10	−0.65	−1.09	41.32	42.11	42.51	0.88*	0.44*
30	5.180e4	6.49	2.72e5	−8.85	+0.02	−0.53	−0.97	40.41	41.20	41.60	1.25*	0.19*
50	5.490e4	9.38	7.18e5	−9.90	+0.13	−0.42	−0.86	39.47	40.26	40.66	1.81*	0.081*
100	5.173e4	18.2	2.14e6	−11.09	+0.02	−0.53	−0.98	38.17	38.96	39.36	3.52*	0.024*
300	4.882e4	40.8	8.51e6	−12.59	−0.09	−0.65	−1.09	36.57	37.35	37.75	7.88*	0.006*
1000	4.807e4	74.8	2.68e7	−13.83	−0.12	−0.67	−1.12	35.29	36.07	36.47	14.44*	0.002*

- If $M \gtrsim 20 M_{\odot}$ Pop III RGB stars have $\mu \gtrsim 10^4$ – 10^5 during caustic transits, they could be detectable for a few months to $\text{AB} \lesssim 29$ mag with JWST.
- Note the combined Bolometric+IGM+K-corrections are more advantageous for Pop III RGB stars and at the higher redshifts ($z \gtrsim 12$)!

Table 4. Implied AGB Pop III Star Observational Parameters Relevant to Caustic Transit Calculations

Mass ^a	T_{eff}^b	Radius ^c	L_{bol}^d	M_{bol}^e	Bolo+IGM+K-corr ^f			AGB m_{UV}^g			t_{rise}^h	transit ⁱ
AGB	— at Helium-depletion —				z=7	z=12	z=17	z=7	z=12	z=17	caust	rate
($M_{\odot}$)	(K)	($R_{\odot}$)	($L_{\odot}$)	(AB)	(AB-mag)			(AB-mag)			(hr)	(/cl/yr)
1.0	6.312e3 ^j	5.23 ^j	39.8 ^j	+0.74	+6.01	+4.57	+4.03	55.99	55.89	56.19	1.01	1.4×10^5
1.5	8.149e3	6.18	151.	−0.71	+3.36	+2.14	+1.64	51.89	52.01	52.35	1.19	4.0×10^3
2.0	1.145e4	4.66	335.	−1.57	+1.06	+0.07	−0.40	48.73	49.08	49.45	0.90	273.
3.0	1.736e4	3.56	1.03e3	−2.79	−0.36	−1.15	−1.60	46.09	46.64	47.03	0.69	28.9
5.0	2.658e4	2.89	3.74e3	−4.19	−0.72	−1.38	−1.82	44.33	45.01	45.41	0.56	6.43
10	3.938e4	3.03	1.98e4	−6.00	−0.42	−1.00	−1.45	42.82	43.57	43.97	0.58	1.71
15	4.215e4	4.55	5.88e4	−7.18	−0.33	−0.90	−1.34	41.73	42.50	42.89	0.88?	0.64?
20	4.386e4	6.14	1.25e5	−8.00	−0.27	−0.84	−1.28	40.97	41.74	42.14	1.19*	0.32*
30	4.006e4	11.7	3.17e5	−9.01	−0.40	−0.98	−1.42	39.83	40.59	40.98	2.26*	0.11*
50	3.536e4	24.3	8.32e5	−10.06	−0.55	−1.15	−1.59	38.63	39.37	39.77	4.70*	0.036*
100	3.392e4	44.0	2.31e6	−11.17	−0.59	−1.19	−1.64	37.49	38.22	38.61	8.50*	0.012*
300	3.165e4	101.	9.19e6	−12.67	−0.64	−1.26	−1.71	35.93	36.65	37.04	19.49*	0.003*
1000	3.122e4	163.	2.26e7	−13.65	−0.65	−1.28	−1.72	34.94	35.66	36.05	31.45*	0.001*

● If $M \gtrsim 20 M_{\odot}$ Pop III AGB stars have $\mu \gtrsim 10^4$ – 10^5 during caustic transits, they could be detectable for a few months to $AB \lesssim 29$ mag with JWST.

● Note the combined Bolometric+IGM+K-corrections are far more advantageous for Pop III AGB stars and at the higher redshifts ($z \gtrsim 12$)!

Table 5. Pop III Stellar Mass Black Hole Accretion Disk Parameters Adopted for Caustic Transit Calculations

Mass ^a	M _{compact} ^b	R _s ^c	Radius ^d	L _{bol} ^e	M _{bol} ^f	bolo+IGM+K-corr ^g			m _{AB} -limits at ^h			t _{rise} ⁱ	Transit ^j
ZAMS		BH	— of the UV accretion disk —			z=7	z=12	z=17	z=7	z=12	z=17	(z=12)	rate
(M _⊙)	(M _⊙)	(km)	(R _⊙)	(L _⊙)	AB-mag	(AB-mag)			(AB-mag)			(hr)	(/cl/yr)
BH accretion-disk bolometric luminosities and UV half-light radii scaling from microlensed quasars (Blackburne et al. 2011)													
30	~5.0 BH	15	1.4	≲4.2×10 ⁴	≳−6.8	−0.6	−1.4	−1.7	≳41.8	≳42.4	≳42.9	0.27?	≳0.58?
50	~24 BH	72	3.0	≲2.0×10 ⁵	≳−8.5	−0.4	−1.2	−1.5	≳40.3	≳40.9	≳41.4	0.58*	≳0.15*
100	~65 BH	195	4.9	≲5.4×10 ⁵	≳−9.6	−0.2	−0.9	−1.3	≳39.4	≳40.0	≳40.5	0.95*	≳0.06*
300	~230 BH	690	9.2	≲1.9×10 ⁶	≳−11.0	−0.2	−1.0	−1.3	≳38.1	≳38.6	≳39.2	1.8*	≳0.02*
1000	~720 BH	2160	16.3	≲6.0×10 ⁶	≳−12.2	−0.2	−0.9	−1.3	≳36.8	≳37.5	≳37.9	3.2*	≳0.01*
BH accretion-disk bolometric luminosities and UV half-light radii estimated from multi-color thin-disk model													
30	~5.0 BH	15	1.9	≲3.1×10 ⁴	≳−6.5	−0.6	−1.4	−1.7	≳42.1	≳42.8	≳43.2	0.37?	≳0.84?
50	~24 BH	72	4.5	≲1.8×10 ⁵	≳−8.4	−0.4	−1.2	−1.5	≳40.4	≳41.1	≳41.5	0.87*	≳0.18*
100	~65 BH	195	7.8	≲5.9×10 ⁵	≳−9.7	−0.2	−0.9	−1.3	≳39.3	≳40.0	≳40.4	1.51*	≳0.06*
300	~230 BH	690	15.8	≲2.0×10 ⁶	≳−11.0	−0.2	−1.0	−1.3	≳38.0	≳38.6	≳39.1	3.1*	≳0.02*
1000	~720 BH	2160	29.8	≲6.6×10 ⁶	≳−12.3	−0.2	−0.9	−1.3	≳36.7	≳37.4	≳37.8	5.8*	≳0.01*

● If $M \gtrsim 20 M_{\odot}$ Pop III stellar mass black hole accretion disks have $\mu \gtrsim 10^4 - 10^5$ during caustic transits, they could be detectable for a few months to $AB \lesssim 29$ mag with JWST. Rise times $\sim$ hours–1 day; Decay times $\lesssim 0.4$ yr.

● Note the combined Bolometric+IGM+K-corrections are also more advantageous for Pop III stellar-mass black hole accretion disks.

Multi- λ model: $T \propto r^{-3/4}$; $T_{max} \simeq 10 \left(\frac{M_{BH}}{100} \right)^{-3/8} \text{ keV}$; $r_{hl} \propto M_{BH}^{1/2}$.